

VESSELS / TANKS	HEAT EXCHANGERS	SAFETY EQUIPMENT	VALVE SYMBOLS	ABBREVIATIONS																																																																												
HORIZONTAL DRUM OR SEPARATOR VERTICAL SEPARATOR / SCRUBBER HORIZONTAL DRUM OR SEPARATOR WITH BOOT FLAT TOP VESSEL FREE STANDING TANK INTEGRATED STORAGE TANK CYCLONE PULSATION DAMPENER VANE PACK DEMISTER (WITH DEFLECTOR DOWNCOMER) CYCLONE DEMISTER MESH PAD DEMISTER BAFFLE VORTEX BREAKER WEIR INLET CYCLONE BOTTLE (NITROGEN OR PROPANE) CYCLONE DEMISTER MESH PAD DEMISTER VANE PACK DEMISTER PACKED SECTION TRAYS CHIMNEY TRAY DIP PIPE MANHOLE NOTE 5	FORCED DRAFT AIR COOLER FORCED DRAFT AIR COOLER WITH LOUVERS CONTROL ELECTRICAL HEATING COIL ELECTRICAL HEATER COMPACT TYPE HEAT EXCHANGER DOUBLE PIPE HEAT EXCHANGER PLATE AND FRAME TYPE EXCHANGER WASTE HEAT RECOVERY UNIT GENERIC SHELL AND TUBE EXCHANGER COIL	SPRINKLER NOZZLE DELUGE SPRAY NOZZLE DELUGE VALVE CO₂ FLOODING SYSTEM INERGEN FLOODING SYSTEM WATER MIST SYSTEM FIRE NOZZLE MONITOR FIXED WATER / FOAM MONITOR HOSE REEL FIRE HYDRANT EYE WASH SAFETY SHOWER WATER CURTAIN	DBB SBB SELF CLOSING VALVE	ABBREVIATIONS <table> <tr> <td>A/S : AIR SUPPLY</td> <td>NC : NORMALLY CLOSED</td> </tr> <tr> <td>ATM : ATMOSPHERE</td> <td>NLL : NORMAL LIQUID LEVEL</td> </tr> <tr> <td>BF : BLIND FLANGE</td> <td>NNF : NORMALLY NO FLOW</td> </tr> <tr> <td>BL : BATTERY LIMIT</td> <td>NO : NORMALLY OPEN</td> </tr> <tr> <td>BP : BIOPROBE</td> <td>OT : OPERATING TEMPERATURE</td> </tr> <tr> <td>CC : CORROSION COUPON</td> <td>PDV : PRODUCTION DIVERTER VALVE</td> </tr> <tr> <td>CM : CORROSION MONITOR</td> <td>PG : PURGE CONNECTION</td> </tr> <tr> <td>CP : CORROSION PROBE</td> <td>PFP : PASSIVE FIRE PROTECTION</td> </tr> <tr> <td>CSC : CAR SEALED CLOSED</td> <td>RB : REDUCED BORE</td> </tr> <tr> <td>CSO : CAR SEALED OPEN</td> <td>ROP : RODDING OUT POINT</td> </tr> <tr> <td>DCV : DIRECTION CONTROL VALVE (SOLENOID VALVE)</td> <td>SA : SAND PROBE</td> </tr> <tr> <td>DBB : DOUBLE BLOCK AND BLEED</td> <td>SB : SWITCHBOARD</td> </tr> <tr> <td>DP : DRAIN POINT</td> <td>SBB : SINGLE BLOCK AND BLEED VALVE</td> </tr> <tr> <td>ECH : ELECTROCHEMICAL / HYDROLYTIC</td> <td>SCC : SCALE / CORROSION COUPON</td> </tr> <tr> <td>EL : ELEVATION</td> <td>SP : SPOOL PIECE/SAMPLING POINT</td> </tr> <tr> <td>EP : EROSION PROBE</td> <td>TEMP : TEMPORARY</td> </tr> <tr> <td>FB : FULL BORE</td> <td>TP : TIE-IN POINT AT PACKAGE BOUNDARY</td> </tr> <tr> <td>FC : FAIL CLOSED</td> <td>TSO : TIGHT SHUT OFF</td> </tr> <tr> <td>FL : FAIL LOCKED</td> <td>T/T : TANGENT LINE TO TANGENT LINE</td> </tr> <tr> <td>FO : FAIL OPEN</td> <td>UC : UTILITY CONNECTION</td> </tr> <tr> <td>FP : FLASHING POINT</td> <td>US : UTILITY STATION</td> </tr> <tr> <td>GRE : GLASS REINFORCED EPOXY</td> <td>UU : UNIT UTILITIES</td> </tr> <tr> <td>HC : HOSE CONNECTION</td> <td>VB : VORTEX</td> </tr> <tr> <td>HH : HAND HOLE</td> <td>VP : VENT POINT</td> </tr> <tr> <td>HP : HIGH PRESSURE</td> <td>VS : VENDOR SUPPLY</td> </tr> <tr> <td>HPT : HIGH POINT</td> <td>XOV : CROSSOVER</td> </tr> <tr> <td>ID : INTERNAL DIAMETER</td> <td></td> </tr> <tr> <td>LC : LOCKED CLOSED</td> <td></td> </tr> <tr> <td>LO : LOCKED OPEN</td> <td></td> </tr> <tr> <td>LP : LOW PRESSURE</td> <td></td> </tr> <tr> <td>LPT : LOW POINT</td> <td></td> </tr> <tr> <td>MAX : MAXIMUM</td> <td></td> </tr> <tr> <td>MCS : MASTER CONTROL STATION</td> <td></td> </tr> <tr> <td>MH : MANHOLE</td> <td></td> </tr> <tr> <td>MIN : MINIMUM LENGTH</td> <td></td> </tr> <tr> <td>MIV : METHANOL INJECTION VALVE</td> <td></td> </tr> <tr> <td>MP : MEDIUM PRESSURE</td> <td></td> </tr> <tr> <td>N : NOZZLE</td> <td></td> </tr> </table>	A/S : AIR SUPPLY	NC : NORMALLY CLOSED	ATM : ATMOSPHERE	NLL : NORMAL LIQUID LEVEL	BF : BLIND FLANGE	NNF : NORMALLY NO FLOW	BL : BATTERY LIMIT	NO : NORMALLY OPEN	BP : BIOPROBE	OT : OPERATING TEMPERATURE	CC : CORROSION COUPON	PDV : PRODUCTION DIVERTER VALVE	CM : CORROSION MONITOR	PG : PURGE CONNECTION	CP : CORROSION PROBE	PFP : PASSIVE FIRE PROTECTION	CSC : CAR SEALED CLOSED	RB : REDUCED BORE	CSO : CAR SEALED OPEN	ROP : RODDING OUT POINT	DCV : DIRECTION CONTROL VALVE (SOLENOID VALVE)	SA : SAND PROBE	DBB : DOUBLE BLOCK AND BLEED	SB : SWITCHBOARD	DP : DRAIN POINT	SBB : SINGLE BLOCK AND BLEED VALVE	ECH : ELECTROCHEMICAL / HYDROLYTIC	SCC : SCALE / CORROSION COUPON	EL : ELEVATION	SP : SPOOL PIECE/SAMPLING POINT	EP : EROSION PROBE	TEMP : TEMPORARY	FB : FULL BORE	TP : TIE-IN POINT AT PACKAGE BOUNDARY	FC : FAIL CLOSED	TSO : TIGHT SHUT OFF	FL : FAIL LOCKED	T/T : TANGENT LINE TO TANGENT LINE	FO : FAIL OPEN	UC : UTILITY CONNECTION	FP : FLASHING POINT	US : UTILITY STATION	GRE : GLASS REINFORCED EPOXY	UU : UNIT UTILITIES	HC : HOSE CONNECTION	VB : VORTEX	HH : HAND HOLE	VP : VENT POINT	HP : HIGH PRESSURE	VS : VENDOR SUPPLY	HPT : HIGH POINT	XOV : CROSSOVER	ID : INTERNAL DIAMETER		LC : LOCKED CLOSED		LO : LOCKED OPEN		LP : LOW PRESSURE		LPT : LOW POINT		MAX : MAXIMUM		MCS : MASTER CONTROL STATION		MH : MANHOLE		MIN : MINIMUM LENGTH		MIV : METHANOL INJECTION VALVE		MP : MEDIUM PRESSURE		N : NOZZLE	
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AIR FILTER BASKET FILTER CARTRIDGE FILTER	COMPRESSORS CENTRIFUGAL COMPRESSOR RECIPROCATING COMPRESSOR SCREW COMPRESSOR BLOWER DRIVERS & TURBINES PNEUMATIC DRIVE GEARBOX VARIABLE SPEED DRIVE DIESEL ENGINE ELECTRIC MOTOR GAS TURBINE PUMPS CENTRIFUGAL PUMP INLINE PUMP GEAR PUMP DIAPHRAGM PUMP ROTARY POSITIVE DISPLACEMENT PUMP VACUUM PUMP VERTICAL PUMP VERTICAL PUMP (SUBMERGED MOTOR) BARREL PUMP PROPORTIONING PUMP RECIPROCATING PUMP SCREW PUMP ECCENTRIC ROTOR PUMP	MISCELLANEOUS PIPE FLARE TIP (WITH PILOTS) SONIC FLARE TIP (WITH PILOTS) MIXER CLEANING MACHINE CALIBRATION POT EJECTOR / HORN DIESEL OIL CENTRIFUGE LABELS TO/FROM DESCRIPTION EQUIPMENT NUMBER DRAWING NUMBER OFF PAGE CONNECTOR CONNECTION TO A PARTICULAR NETWORK TIE-IN HOOK UP CONTINUATION LINE CUT SYSTEM BOUNDARY PROCESS LINES INDICATION OF REVISION CHANGE MAJOR LINE (FLOW DIRECTION INDICATED) HEAT TRACING LINE FLEXIBLE LINE MINOR LINE SKID OUTLINE (LIMIT OF PACKAGE SUPPLY SHOWING TERMINAL POINTS) FUTURE LINE	PIPING SYMBOLS CAP (WELDED) PIPING CLASS SPECIFICATION BREAK OR INSULATION CHANGE CLAMP ASSEMBLY EXPANSION BELLOW SYPHON FLAME ARRESTOR (INLINE OR END OF LINE) VENT (GOOSE NECK) VENT WITH HOOD BIRD SCREEN STATIC MIXER SAND DETECTOR CORROSION / EROSION PROBE WITH ACCESS FITTINGS X=C/E CORROSION COUPON WITH ACCESS FITTING SAMPLE POINT BIOPROBE WITH ACCESS FITTING SILENCER SPADE SPACER SPECTACLE BLIND (CLOSED) SPECTACLE BLIND (OPEN) FEMALE HOSE CONNECTION MALE HOSE CONNECTION HOSE CONNECTION WITH PLUG THREADED PLUG REDUCER (SHOWING LINE SIZE CHANGE) CONTINUOUS SLOPE (WITH SLOPE VALUE) NOTE 1 GRAVITY - DRAINED LINE - (NO POCKETS) (NOTE 4) SPECIAL PIPING ITEM SPOOL PIECE BARRED TEE LIQUID TRAP DRAIN QUICK COUPLING FLEXIBLE HOSE OPEN DRAIN TUNDISH DRAIN BOX SEAL POT FLOW STRAIGHTENER RESTRICTION ORIFICE TANK RUNDOWN PIPE SEAL (GOOSE NECK) BLEED RING (WITH BALL VALVE AS BLEED) FLEXIBLE HOSE WITH BREAK-AWAY COUPLING FLEXIBLE HOSE WITH BOWSER TEMPORARY STRAINER Y-TYPE STRAINER T-TYPE STRAINER CONICAL TYPE STRAINER VENT TIP	NOTES: 1 - 1% SLOPE FOR DRAINAGE (GS-SAF-228 § 4.4.2) 0.2% SLOPE FOR FLARE (GS-ECP-103 § 13.8.1) 2 - TO BE INSTALLED ONLY ON PIPING ACCESSIBLE TO OPERATORS. 3 - ALL BALL VALVES 1.5" AND BELOW ARE FULL BORE. 4 - SPECIFY "NO POCKETS" AS APPLICABLE / REQUIRED. 5 - NOZZLE DESIGNATION : A. MAIN PRODUCT INLET B. MAIN PRODUCT OUTLET C. CONDENSATE/OIL OUTLET D. DRAIN E. WATER INLET, OUTLET F. VENT G. LEVEL GAUGE H. OVERFLOW J. AUXILIARY PRODUCT INLET K. AUXILIARY PRODUCT OUTLET L. LEVEL INSTRUMENT M. MANHOLE N. HANDHOLE-NOZZLE SUPPORT P. PRESSURE CONNECTION R. TO / FROM HEAT EXCHANGER S. PRESSURE SAFETY VALVE CONNECTION T. TEMPERATURE CONNECTION U. UTILITY CONNECTION-SAMPLE V. STREAM INLET W. SPARE CONNECTION WHEN IN THE PRESENCE OF SEVERAL CONNECTIONS OF SAME MARK, INDICATE AS FOLLOWS : A1,A2,..An. REF. : GS-PVV-202 P&ID LEGENDS AND TYPICALS GENERAL SYMBOLS																																																																												
FILTERS / STRAINERS AIR FILTER BASKET FILTER CARTRIDGE FILTER	PUMPS VERTICAL PUMP VERTICAL PUMP (SUBMERGED MOTOR) BARREL PUMP PROPORTIONING PUMP RECIPROCATING PUMP SCREW PUMP ECCENTRIC ROTOR PUMP	VALVE SYMBOLS FLANGED VALVE (TYPICAL OF ALL VALVE TYPES) FLANGED & WELDED VALVE RATED CHECK VALVE (SPRING LOADED) WELDED VALVE (TYPICAL OF ALL VALVE TYPES) GATE VALVE BALL VALVE FULL BORE (NOTE 3) BALL VALVE REDUCED BORE GLOBE VALVE ANGLE VALVE NEEDLE VALVE BUTTERFLY VALVE PLUG VALVE MANUALLY OPERATED CHOKE VALVE REMOVEDLY OPERATED CHOKE VALVE MINIMUM FLOW VALVE (AUTOMATIC RECIRCULATION VALVE) VACUUM AIR RELEASE VALVE BREATHER VALVE 3-WAY VALVE DUAL PLATE (WAFER) CHECK VALVE CHECK VALVE SWING (FLANGED) CHECK VALVE DIAPHRAGM VALVE HAND ANGLE VALVE INJECTION RATE CONTROL DEVICE (IRCD) FLOAT VALVE	INSULATION INSULATED LINE (INSULATION CLASS X) INSULATED EQUIPMENT (INSULATION CLASS X) INSULATION CLASSES: PB: A PHYSICAL BARRIER TO BE APPLIED TO PIPING SYSTEMS OPERATING ABOVE 70°C IN AREA WHERE THE PIPE WORK IS ACCESSIBLE TO PERSONNEL. (NOTE 2) P : PERSONNEL PROTECTION (NOTE 2) H : HEAT CONSERVATION E : ELECTRICAL TRACING C : COLD CONSERVATION A : ACOUSTIC INSULATION B : SOLAR INSULATION D : DUAL (COLD AND HOT) I : ANTI ICING K : ANTI CONDENSATION L : COLD AND ACOUSTIC R : LIQUID TRACING S : STREAM TRACING T : HOT AND ACOUSTIC																																																																													

INSTRUMENT SYMBOLS

INSTRUMENT LINE SYMBOLS		COMPLEMENTARY FUNCTIONS		INSTRUMENTS	
CONNECTION TO PROCESS LINE OR INSTRUMENT SUPPLY CAPILLARY TUBE ELECTRIC SIGNAL PNEUMATIC SIGNAL HYDRAULIC SIGNAL ELECTROMAGNETIC OR SONIC SIGNAL (GUIDED) ELECTROMAGNETIC OR SONIC SIGNAL (NOT GUIDED) SERIAL LINK (SOFTWARE OR DATA LINK) MECHANICAL LINK UNDEFINED SIGNAL ELECTRIC SIGNAL TEMPORISED INSTRUMENT TAG NUMBERING SYSTEM NN - AAAAAA - NN - NN - NN INSTR. LOOP NUMBER (NOT FOR WELLS/GASLIFT) MECH. SEQUENTIAL NUMBER SYSTEM / SUB SYSTEM NUMBER INSTRUMENT IDENTIFICATION (2-6 LETTERS) PLATFORM NUMBER INSTRUMENT TAG PRESENTATION ON PID's AAAAAA INSTRUMENT IDENTIFICATION (2-6 LETTERS) NN ZZ SEQUELIEL NUMBER EQUIPMENT / ELEVATION DECK OR WELL SEQUELIEL NUMBER SYSTEM / SUBSYSTEM CODE FOR MORE DETAILS REFER TO OFPG-PP-PMT-PGM-012		SUMMING THE OUTPUT EQUALS THE ALGEBRAIC SUM OF THE INPUTS (THE INPUTS MAY BE LABELLED WITH POSITIVE OR NEGATIVE SIGNS). AVERAGING THE OUTPUT EQUALS THE ALGEBRAIC SUM OF THE INPUTS DIVIDED BY THE QUANTITY OF INPUTS. DIFFERENCE THE OUTPUT EQUALS THE ALGEBRAIC DIFFERENCE OF THE INPUTS. PROPORTIONAL THE OUTPUT IS DIRECTLY PROPORTIONAL TO THE INPUT. IN THE CASE OF A VOLUME BOOSTER, "K" MAY BE REPLACED BY 1:1 FOR FIXED GAINS, 2:1, 3:1, ETC MAY BE SUBSTITUTED FOR K. INTEGRAL THE OUTPUT VARIES IN ACCORDANCE WITH BOTH MAGNITUDE AND DURATION OF THE INPUT. THE OUTPUT IS PROPORTIONAL TO THE TIME INTEGRAL OF THE INPUTS. DERIVATIVE THE OUTPUT IS PROPORTIONAL TO THE RATE OF CHANGE (DERIVATIVE) OF THE INPUT. MULTIPLYING THE OUTPUT EQUALS THE PRODUCT OF THE INPUTS. DIVIDING THE OUTPUT EQUALS THE QUOTIENT OF THE 2 INPUTS. ROOT EXTRACTION THE OUTPUT EQUALS THE ROOT (I.E. CUBE ROOT , FOURTH ROOT, 3/2 ROOT, ETC.) OF THE INPUTS. IF n IS OMITTED, A SQUARE ROOT IS ASSUMED. EXPONENTIAL THE OUTPUT EQUALS THE INPUT RAISED TO A POWER (I.E. SECOND, THIRD, FOURTH, ETC.). NON LINEAR OR UNSPECIFIED FUNCTION THE OUTPUT EQUALS SOME NONLINEAR OR UNSPECIFIED FUNCTION OF THE INPUT. TIME FUNCTION THE OUTPUT EQUALS THE INPUT TIMES SOME FUNCTION OF TIME OR EQUALS SOME FUNCTION OF TIME ALONE. HIGH SELECTING THE OUTPUT IS EQUAL TO THE GREATER OF THE INPUTS. LOW SELECTING THE OUTPUT IS EQUAL TO THE LESSER OF THE INPUTS. HIGH LIMITING THE OUTPUT EQUALS THE INPUT OR THE HIGH LIMIT VALUE WHICHEVER IS LOWER. LOW LIMITING THE OUTPUT EQUALS THE INPUT OR THE LOW LIMIT VALUE WHICHEVER IS HIGHER. REVERSE PROPORTIONAL THE OUTPUT IS REVERSELY PROPORTIONAL TO THE INPUT. VELOCITY LIMITER THE OUTPUT EQUALS THE INPUT AS LONG AS THE RATE OF CHANGE OF THE INPUT DOES NOT EXCEED A LIMIT VALUE. THE OUTPUT WILL CHANGE AT THE RATE ESTABLISHED BY THIS LIMIT UNTIL THE OUTPUT AGAIN EQUALS THE INPUT. BIAS THE OUTPUT EQUALS THE INPUT PLUS (OR MINUS) SOME ARBITRARY VALUE (BIAS). CONVERT THE FORM OF THE OUTPUT SIGNAL IS DIFFERENT FROM THAT OF THE INPUT : * E : VOLTAGE I : CURRENT P : PNEUMATIC A : ANALOG B : BINARY * H : HYDRAULIC O : ELECTROMAGNETIC, SONIC R : RESISTANCE (ELECTRIC) D : DIGITAL X Y Y X X : THE INPUT SIGNAL PRODUCES ALARM THRESHOLDS (H, L) Y : THE INPUT SIGNAL PRODUCES CONTROL THRESHOLDS (ON/OFF, START/STOP) CONTROL VALVE HAND-OPERATED ONLY CONTROL VALVE WITH DIAPHRAGM ACTUATOR CONTROL VALVE WITH DIAPHRAGM ACTUATOR AND MANUAL OPERATOR CONTROL VALVE WITH DIAPHRAGM ACTUATOR AND POSITIONER AND MANUAL OPERATOR. CONTROL VALVE WITH DIAPHRAGM ACTUATOR AND TRAVEL STOP (SHOWN AS MINIMUM STOP) VALVE WITH ELECTRIC ACTUATOR (GENERAL) 2-WAY SOLENOID VALVE 2-WAY SOLENOID VALVE WITH REMOTE RESET 3-WAY SOLENOID VALVE (SHOWING OPEN PATH WHEN DE-ENERGISED) SELF-CONTAINED PRESSURE REDUCING REGULATOR SELF-CONTAINED BACK-PRESSURE REGULATOR VALVE WITH CYLINDER ACTUATOR SINGLE ACTING VALVE WITH CYLINDER ACTUATOR DOUBLE ACTING CONVENTIONAL PRESSURE OR VACUUM RELIEF VALVE PILOT OPERATED RELIEF VALVE QUICK OPENING VENT VALVE RUPTURE DISC PRESSURE AND VACUUM RELIEF MANHOLE COVER ORIFICE PLATE BETWEEN FLANGES VENTURI TYPE FLOWMETER ORIFICE PLATE WITH QUICK CHANGE FITTING PITOT TUBE OR ANNUBAR CORIOLIS FLOW METER POSITIVE DISPLACEMENT METER TURBINE TYPE FLOWMETER VORTEX TYPE FLOWMETER ELECTROMAGNETIC TYPE FLOWMETER MASS FLOWMETER FLOW INDICATOR ROTAMETER VORTEX INSERTION TYPE ULTRASONIC TYPE FIELD MOUNTED INSTRUMENT MAIN CONTROL ROOM MOUNTED INSTRUMENT INACCESSIBLE OR BACK OF PANEL MOUNTED INSTRUMENT INSTRUMENT MOUNTED IN AUXILIARY LOCATION NORMALLY ACCESSIBLE TO OPERATOR INSTRUMENT MOUNTED BEHIND LOCAL PANEL SAFETY CONTROL FUNCTION SYSTEM (PSS, FGS, ESD) PLC SAFETY CONTROL FUNCTION SYSTEM (PLC) PLC PROCESS CONTROL FUNCTION PACKAGE SYSTEM PROCESS CONTROL FUNCTION (MAIN CONTROL ROOM) PROCESS CONTROL FUNCTION (INACCESSIBLE OR BACK OF PANEL MOUNTED INSTRUMENT) PROCESS CONTROL FUNCTION (INSTRUMENT MOUNTED IN AUXILIARY LOCATION) INTERLOCK LOGIC X = LOGIC INPUT X = LOGIC OUTPUT PCS: PROCESS CONTROL SYSTEM PSS: PROCESS SAFETY SYSTEM ESD: EMERGENCY SHUTDOWN SYSTEM FGS: FIRE & GAS SYSTEM HIPS: HIGH INTEGRITY PROTECTION SYSTEM 3-WAY SWITCH (2 POSITIONS) 4-WAY SWITCH FIELD MOUNTED INSTRUMENT WITH DUAL FUNCTION LIGHT / FLAG SUPPLEMENTARY SYMBOLS FCP FCP: FIRE CONTROL PANEL # #: LOCATION TYPES CAN BE COMBINED WITH THE DIFFERENT FUNCTIONS + CONTROL SYSTEM TYPES CAN BE COMBINED WITH DIFFERENT FUNCTIONS IF + IS OMITTED, THEN PCS IS ASSUMED.			

P&ID LEGENDS AND TYPICALS GENERAL SYMBOLS

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